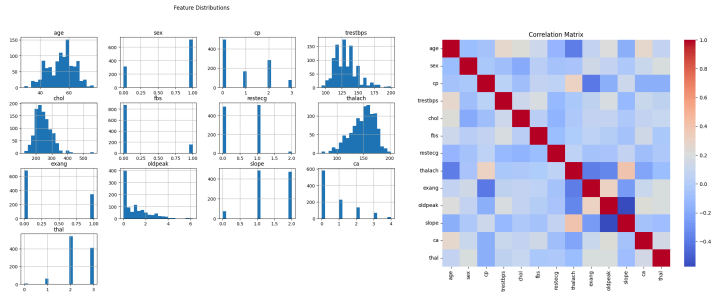
Data & Preprocessing

The heart disease dataset contains 1,025 samples across 13 clinical features with no missing values. The target variable is binary (no disease vs. disease), with a near-balanced class distribution of 500 vs. 525 samples. Features span a mix of continuous variables (age, resting blood pressure, cholesterol, maximum heart rate, ST depression) and categorical/ordinal ones (chest pain type, resting ECG, thalassemia type, etc.). Feature distributions are shown.



The correlation matrix reveals that chest pain type, maximum heart rate, number of vessels, and thalassemia type show the strongest associations with the target. Preprocessing followed a leakage-free pipeline of mean imputation → standard scaling → SVM. All steps were fit on training data only and applied to the test set. The data was split 80/20, yielding 820 training and 205 test samples. The aging dataset contains DNA methylation beta values for 37 samples across 485,515 CpG site features. This is a severely high-dimensional, low-sample-count setting. Missing values (526 total) were handled via column-wise mean imputation without dropping any rows or columns. The regression target is chronological age, ranging from 65 to 94 years (mean: 73). Given the extreme dimensionality, PCA was applied within the pipeline prior to SVR, reducing the feature space to 50 components. The data was split 80/20, with 29 training and 8 test samples. The very small test set (n=8) can lead to prediction errors that carry disproportionate weight, and results should be interpreted accordingly.

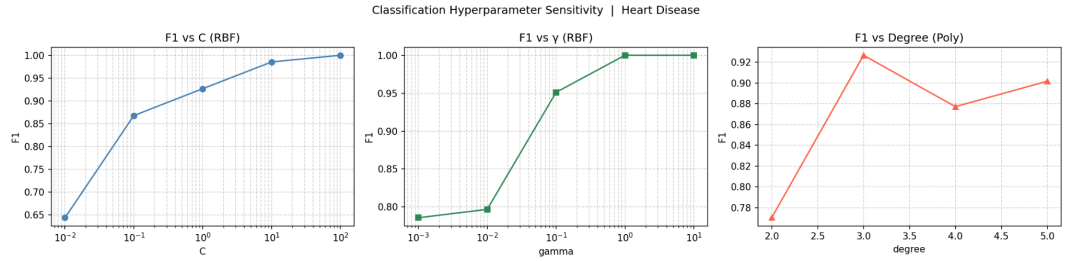
Modeling & Hyperparameters

Three SVM kernels were evaluated: linear, polynomial, and RBF. Each was trained on the standardized feature set with default regularization (C=1.0) as a baseline, then tuned across the hyperparameters most relevant to each kernel.

```
--- Kernel Experiments (Classification) ---
  kernel=linear | acc=0.8146 | f1=0.8120 | prec=0.8290 | rec=0.8146
  kernel=poly   degree=2 | acc=0.7707 | f1=0.7703 | prec=0.7716 | rec=0.7707
  kernel=poly   degree=3 | acc=0.9268 | f1=0.9266 | prec=0.9300 | rec=0.9268
  kernel=poly   degree=4 | acc=0.8780 | f1=0.8772 | prec=0.8859 | rec=0.8780
  kernel=rbf    gamma=0.01 | acc=0.8000 | f1=0.7969 | prec=0.8149 | rec=0.8000
  kernel=rbf    gamma=0.1 | acc=0.9512 | f1=0.9512 | prec=0.9519 | rec=0.9512
  kernel=rbf    | acc=0.9268 | f1=0.9268 | prec=0.9271 | rec=0.9268

Best config: {'kernel': 'rbf', 'C': 1.0, 'gamma': 0.1}  (F1 = 0.9512)
```

The linear kernel achieved F1 = 0.81, serving as a reasonable baseline given the mix of continuous and categorical features. Polynomial kernels improved substantially at degree 3 (F1 = 0.93) but degraded at degree 4, suggesting mild overfitting to training geometry. RBF performed best overall, with the optimal configuration at C=1.0 and γ=0.1 achieving F1 = 0.95.



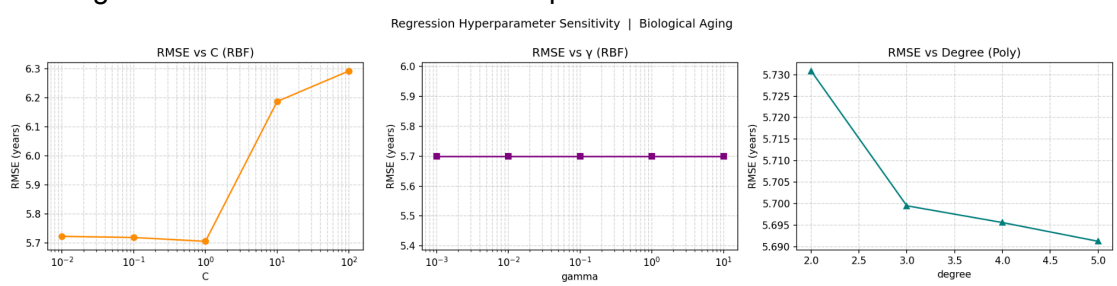
Increasing C monotonically improved RBF performance up to around C=10, beyond which gains plateaued. gamma showed a sharp elbow between 0.1 and 1.0, where the kernel transitions from underfitting (too smooth) to near-perfect separation on training data. The polynomial degree peaked at 3 and declined thereafter.

The same three kernel families were evaluated for SVR, with PCA-50 applied upstream in all pipelines. Results are shown.

```
--- Kernel Experiments (Regression) ---
  kernel=linear | MAE=5.826 | RMSE=6.465 | R²=-0.2868
  kernel=poly   degree=2 | MAE=4.143 | RMSE=5.731 | R²=-0.0110
  kernel=poly   degree=3 | MAE=4.115 | RMSE=5.700 | R²=-0.0000
  kernel=poly   degree=4 | MAE=4.114 | RMSE=5.696 | R²=0.0014
  kernel=rbf    gamma=0.01 | MAE=4.238 | RMSE=5.700 | R²=-0.0003
  kernel=rbf    gamma=0.1 | MAE=4.238 | RMSE=5.700 | R²=-0.0003
  kernel=rbf    | MAE=4.239 | RMSE=5.706 | R²=-0.0023

Best config: {'kernel': 'poly', 'C': 1.0, 'degree': 4, 'gamma': 'scale'}  (R² = 0.0014)
```

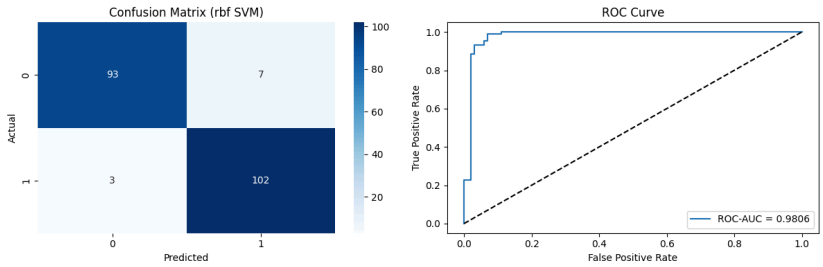
The linear kernel performed worst ($R^2 = -0.29$), indicating it failed to capture even basic variance in the target. RBF performance was essentially flat across all values of gamma, as the RMSE vs. γ plot is a horizontal line, which is a strong indicator that the kernel could not find meaningful structure in the PCA-reduced space.



Polynomial kernels outperformed both, with degree 4 achieving the best result (MAE = 4.11, RMSE = 5.70, $R^2 = 0.0014$). The near-zero R^2 across all configurations reflects the fundamental challenge of this dataset rather than a modeling failure: with only 29 training samples and 485,515 original features, no kernel had sufficient data to learn a generalizable mapping from methylation patterns to age. The SVR predictions converge to a near-constant output close to the training mean (~69.5 years), which is consistent with a model that has learned the marginal distribution of the target but little else.

Metrics & Visuals

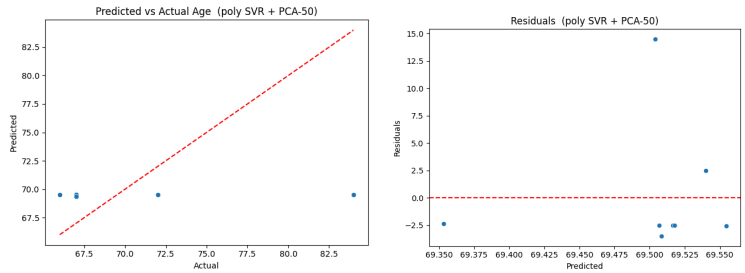
```
Final metrics — Accuracy: 0.9512 | F1: 0.9512 | ROC-AUC: 0.9806
```



The confusion matrix shows 93 true negatives and 102 true positives, with only 10 total misclassifications (7 false positives, 3 false negatives). The ROC curve confirms strong discriminative ability with AUC = 0.98, indicating the model reliably separates positive and negative cases across all classification thresholds. The curve rises steeply toward the top-left corner, reflecting high sensitivity even at low false positive rates.
The final polynomial SVR (degree 4, PCA-50) achieved the following on the 8-sample test set

```
Final metrics — MAE: 4.114 | RMSE: 5.696 | R²: 0.0014
```

R^2 = 0.0014 reveals that the model explains virtually none of the variance in age. It performs only marginally better than simply predicting the mean.



The parity plot illustrates that predictions cluster tightly between 69.35 and 69.56 years regardless of actual age, which ranges from 66 to 84 in the test set. Points should scatter along the diagonal for a well-fitted model, but instead they form a near-horizontal band, confirming the model has collapsed to a mean-prediction strategy. The residual plot is consistent with this, as the residuals show no structure around zero and scale with the distance of the true age from the predicted mean.

Sensitivity (reference sensitivity graphs above)

Among hyperparameters, C had the largest impact, it improved performance steadily up to C=10 before saturating. Gamma showed a sharp elbow around 0.1–1.0, and polynomial degree peaked at 3 before declining. For PCA, 50 components was a practical constraint, not an optimized one, and likely discards age-relevant signals. C showed a non-monotonic effect, with RMSE rising sharply above C=1. Gamma had no detectable effect across all tested values, consistent with RBF failing entirely on this task. Polynomial degree produced the only meaningful sensitivity signal, with RMSE improving modestly from degree 2 to 5.